Explanation of block diagram of digital energy meter

The digital meter has a power supply, metering circuit, a processing and communication circuits containing a microcontroller and real time clock circuit as shown in Figure 10.29 (b).

It has also current sensor and voltage sensor circuits to measure the current and voltage consumed by the users. Power is calculated by multiplying voltage and current. The time duration of consumption is multiplied with the voltage and current, in order to calculate the power consumption in kilowatt hour (kwh) and it is commercially called it as ' Unit'. It is periodically stored in memory circuit from time to time and added with the previously stored consumed values. We can see often, the latest number of units consumed, in the display.

LEARNING OUTCOMES

A student will understand the working principle of the following instruments after reading this Chapter.

1. **Multimeter (Analog & Digital)**
2. **Cathode Ray Oscilloscope (C.R.O.)**
3. **Signal Generator**
4. **Function Generator**
5. **Spectrum analyser**
6. **Logic probe**
7. **IC tester**
8. **Digital energy meter**

GLOSSARY

S. No	Terms	Explanation
1	**Function generator**	Signal generator that can produce sine, triangle and sawtooth output waveforms
2	**Ohm meter**	An instrument which is used for measuring ohms.
3	**Cathode Ray oscilloscope**	It is an electronic device in which various types of waveforms can be seen on the screen.
4	**Digital Meter**	An instrument that uses counting methods
5	**Calibration**	To adjust the correct value of a reading by comparison to a standard
6	**Continuity**	Occurs when a complete path for current exists
7	**Multimeter**	Electronic test equipment that can perform multiple tasks. Typical one is capable of measuring voltage, current and resistance.
8	**Spectrum analyser**	Instrument used to display the frequency domain of a waveform by plotting amplitude against frequency
9	**Energy meter**	Instrument used to measure electrical energy in units
10	**Voltmeter**	Instrument used to measure difference in potential between two points

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